

# CAPITAL ALLOCATION REQUEST — INVESTMENT MEMO

**To:** Fund Investment / Risk Committee **From:** K. Nguyen — Execution & Systematic Strategies **Re:** Capital allocation to the **Market-Neutral Pairs + Momentum Blend** (the "Book") **Date:** 2026-06-07  
**Classification:** Internal — Capital Committee

**THE ASK IN ONE LINE.** Allocate **\$35M** to a dollar-neutral, market-neutral systematic equity book (OOS Sharpe **1.73**, vol **3.2%**, market beta **-0.02**, max drawdown **-1.9%**). \$35M is **65% of measured capacity (\$53.6M)** — the point at which our own market impact begins to erode the edge — leaving deliberate headroom. Expected net contribution: **\$1.2–1.7M/yr** at a net Sharpe of **1.1–1.6**.

## 1. Strategy Thesis — what we are funding

The Book combines **two structurally uncorrelated systematic equity strategies** into a single risk-optimized, dollar-neutral portfolio:

- **Cointegration Pairs (mean-reversion), 80% of risk.** A basket of 9 within-sector pairs — not a single fragile relationship — each gated on three independent tests: Engle–Granger cointegration  $p$ , ADF stationarity of the spread, and a mean-reversion half-life between 1 and 90 days. Positions enter at  $\pm 1.5\sigma$ , exit at  $0.5\sigma$ , hard-stop at  $3.5\sigma$ ; legs are dollar-neutral; the basket is inverse-vol weighted.
- **Cross-sectional Momentum (trend), 20% of risk.** 60-1 formation (trailing 60-day return, skipping the last 5 days), long top quintile / short bottom quintile, equal-weight, monthly rebalance.

**Why blend them.** The two sleeves are nearly uncorrelated ( $\rho = -0.11$  in calm markets). Mean-variance optimization with **Ledoit–Wolf shrinkage** covariance places **~80/20** in pairs/momentum at the minimum-variance point. The result is a book whose realized **market beta is -0.02** — a genuine *diversifier*, not a repackaged long.

### The 9-pair basket (calibrated 2022):

| Pair      | Sector      | Coint $p$ | Half-life (d) | Wt    |
|-----------|-------------|-----------|---------------|-------|
| PEP/MCD   | Staples     | 0.002     | 4.5           | 16.2% |
| KO/CL     | Staples     | 0.008     | 6.9           | 16.3% |
| MS/BLK    | Financials  | 0.032     | 10.7          | 9.5%  |
| SPY/GLD   | Cross-asset | 0.052     | 8.8           | 9.5%  |
| MSFT/ADBE | Tech        | 0.054     | 7.5           | 7.1%  |
| JNJ/PFE   | Health      | 0.056     | 8.3           | 11.1% |

| Pair    | Sector      | Coint $p$ | Half-life (d) | Wt    |
|---------|-------------|-----------|---------------|-------|
| V/MA    | Payments    | 0.058     | 7.6           | 17.9% |
| UNH/MRK | Health      | 0.064     | 8.5           | 6.1%  |
| GE/LOW  | Industrials | 0.072     | 8.0           | 6.2%  |

**Honest scope note (why this and not an intraday book).** Prior research built an intraday order-book residual stat-arb signal that *looked* strong (Sharpe  $\sim 1.6$ ) but was **retracted under our own kill-test**: it was bid-ask bounce that dies on a one-bar execution lag. We do **not** ask for capital on it. Instead we repurpose that order-book work as the **market-impact calibration engine** behind §4. Funding the strategy that survived honest out-of-sample validation — and disclosing the one that didn't — is the discipline this committee should expect.

## 2. Backtest Performance — the honest scorecard

All figures below are **out-of-sample** (expanding-window, purged walk-forward, 21-day rebalance, Ledoit–Wolf shrinkage, turnover-damped, net of cost). OOS window **2022-06-09 → 2022-12-30 (142 trading days)**. Selection used only pre-window data.

| Book                                      | OOS Sharpe  | Ann. Return | Ann. Vol    |
|-------------------------------------------|-------------|-------------|-------------|
| <b>Robust walk-forward Blend (funded)</b> | <b>1.73</b> | <b>5.5%</b> | <b>3.2%</b> |
| 100% Pairs basket                         | 5.36        | 13.0%       | 2.4%        |
| 100% Momentum                             | −1.82       | −21.1%      | 11.6%       |
| Static 60/40                              | −0.13       | −0.6%       | 4.8%        |

See [figures/walkforward\\_oos.png](#) (carried from the strategy build) and [figures/drawdown.png](#).

**Reading this honestly.** In choppy 2022, momentum was a structural loser; the robust optimizer *correctly downweighted it*, and the Blend **crushed naive 60/40 and beat the weak sleeve**. It did **not** beat pairs-alone — and we do not claim it should. The pairs-alone 5.36 Sharpe is an in-sample-flavored, frictionless figure treated as a *ranking signal, not a promise*. **The disciplined claim is this:** allocation protects the book from its weak sleeve and delivers a low-volatility, market-neutral return stream — it does not manufacture alpha out of one sleeve.

**Data note.** Strategy returns are calibrated on 2022 daily closes (the OHLC sample available); the order-book record runs to 2026-03 and is used for the forward liquidity and impact picture in §4. The short return sample is a known limitation and is the reason the ask is sized conservatively and staged (§5).

## 3. Risk Metrics

Computed on the OOS net daily P&L of the Blend (142 days):

| Metric            | Value       | Read                               |
|-------------------|-------------|------------------------------------|
| Annualized Sharpe | <b>1.73</b> | risk-adjusted return               |
| Sortino           | <b>2.88</b> | downside-only; left tail is benign |

| Metric                       | Value           | Read                                      |
|------------------------------|-----------------|-------------------------------------------|
| Max Drawdown                 | <b>-1.95%</b>   | shallow; recovers quickly                 |
| Calmar                       | <b>2.82</b>     | return per unit of worst loss             |
| Daily VaR (95% / 99%)        | -0.31% / -0.42% | risk-limit anchors                        |
| Daily CVaR (95% / 99%)       | -0.39% / -0.52% | expected tail loss                        |
| Skew / Excess kurtosis       | +0.08 / +0.74   | <b>not</b> "pennies before a steamroller" |
| <b>Market beta</b>           | <b>-0.02</b>    | <b>proves market-neutrality</b>           |
| Probabilistic Sharpe (vs 0)  | 0.90            | 90% confidence Sharpe > 0                 |
| Deflated Sharpe (100 trials) | 0.11            | <i>see caveat</i>                         |

**The one number we will not hide — Deflated Sharpe = 0.11.** After deflating for multiple testing ( $\approx 100$  pair-search trials) over only 142 OOS days, the Sharpe is **not yet statistically distinguishable** from what a search of that size could produce by luck (expected max Sharpe under the null  $\approx 3.4$  given the tiny sample). This is a *sample-length* limitation, not a sign the edge is fake — the economics (cointegration, sector structure, near-zero beta) are sound. **Mitigant:** we size the ask well below capacity and **stage** deployment (§5) so live data extends the track record before full size is committed. We would rather state this openly than be caught by it.

**Stress — 1987 Black Monday overlay.** Replaying the Book against the 1987 crash tape (S&P futures **-21.2%** worst day, **-28.6%** peak-to-trough;  $\sim 7.8\times$  normal daily vol):

| Channel                                                         | Modeled 1-day impact                           |
|-----------------------------------------------------------------|------------------------------------------------|
| Directional ( $\beta \times$ crash)                             | <b>+0.5%</b> (slight short beta <i>helps</i> ) |
| Idiosyncratic / correlation-breakdown (99%, at $7.8\times$ vol) | <b>-3.6%</b>                                   |
| <b>Net scenario loss</b>                                        | <b>-3.1%</b>                                   |
| <b>After -5% kill-switch</b>                                    | <b>-3.1% (inside the stop)</b>                 |

Even a Black-Monday-scale shock produces a modeled  $\sim -3\%$  **day**, contained by the governance limits in §5. The book's only real crash vulnerability is correlation breakdown (sleeve  $\rho$  moves  $-0.11 \rightarrow +0.24$  on worst-decile days), which we already size to — the headline weight is set to the *stressed* correlation, not the calm one.

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## 4. Capacity Analysis — "How much can we trade before we ruin the spread?"

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This is the core of the request. Capacity is set by the point where **our own market impact eats the gross edge**. We calibrate impact directly from the **L1–L3 order book** (213M quote rows; 50 traded names; 2022, matching the backtest window).

**Step 1 — Liquidity calibration (per name, from the book).** Median half-spread **4.24 bps**; displayed L1–L3 depth (ETFs deepest: SPY  $\approx$  \$100k/snapshot, single names  $\approx$  \$25k); annualized mid volatility; and **Kyle's  $\lambda$**  (price impact per share), estimated by regressing minute mid-changes on **order-flow imbalance** (Cont–Kukanov–Stoikov). *Note:* the feed carries quotes only (no executed volume), so

all impact estimates are quote-based and, where displayed depth is used as the liquidity scale, deliberately **conservative**.

**Step 2 — Cost vs. capital.** The Book's **gross alpha is 5.66%/yr** at a one-way **turnover of 15.3× book/yr**. Crossing the spread is size-independent and costs **0.67%/yr**. Market impact *grows with capital*. We price it two ways:

- **Kyle's  $\lambda$  (linear, directly calibrated):** the *optimistic* bound — permanent impact is gentle; net Sharpe stays above the hurdle past **\$3B**.
- **Square-root law (Almgren) on displayed depth:** the *conservative* bound — net Sharpe falls to the **1.0 hurdle at  $\approx$  \$54M**.

These **bracket** true capacity. Per Jane-Street discipline, **we anchor on the conservative bound**.

**Step 3 — The capacity frontier**, calibrated on two liquidity regimes

( `figures/capacity_frontier.png` , `figures/capacity_frontier_full.png` ):

| Capacity definition                                                                            | 2022 (stressed yr)  | 2022–2026 (forward) |
|------------------------------------------------------------------------------------------------|---------------------|---------------------|
| <b>Hard capacity — <math>\sqrt{\text{-law}}</math> + Sharpe <math>\geq 1.0</math> (anchor)</b> | <b>\$53.6M</b>      | <b>\$76.2M</b>      |
| Participation cap — $\leq 10\%$ of displayed depth                                             | \$113.7M            | \$98.7M             |
| Hard capacity — Kyle- $\lambda$ (optimistic bound)                                             | $> \$3,000\text{M}$ | $> \$3,000\text{M}$ |
| Median half-spread                                                                             | 4.24 bps            | 3.75 bps            |

**STATED CAPACITY = \$53.6M (stressed) to \$76.2M (forward).** We **anchor on the stressed-2022 number, \$53.6M** — the most conservative binding constraint ( $\sqrt{\text{-law}}$  + hurdle), well inside the participation cap and far inside the Kyle- $\lambda$  bound. The forward, full-period liquidity (tighter spreads, gentler impact as markets normalized after 2022) lifts capacity to **\$76.2M**, which we treat as the **scale-up target**, not the opening ask. Beyond capacity, net Sharpe drops below the Board's 1.0 floor.

`figures/impact_curve.png` shows net dollar P&L vs. capital — it rises, then rolls over as impact compounds: the visual answer to "how much before we ruin the spread."

**Capacity is hurdle-sensitive** ( $\sqrt{\text{-law}}$ , forward period) — the Board's chosen Sharpe floor materially moves the number:

| Board hurdle (Sharpe) | 0.50   | 0.75   | 1.00         | 1.25  | 1.50 |
|-----------------------|--------|--------|--------------|-------|------|
| Stated capacity       | \$251M | \$151M | <b>\$76M</b> | \$26M | \$6M |

We adopt the **1.00 hurdle** as the prudent default; a softer hurdle would justify a materially larger allocation.

## 5. The Capital Request

**We request \$35M — 65% of the \$53.6M stated capacity**, leaving **\$18.6M of headroom** by design.

**Expected economics at \$35M** (impact priced at the requested size):

|                                 | Conservative ( $\sqrt{\lambda}$ -law) | Optimistic (Kyle- $\lambda$ ) |
|---------------------------------|---------------------------------------|-------------------------------|
| Net annual return               | 3.5%                                  | 5.0%                          |
| <b>Net Sharpe</b>               | <b>1.11</b>                           | <b>1.56</b>                   |
| <b>Net P&amp;L contribution</b> | <b>\$1.23M/yr</b>                     | <b>\$1.74M/yr</b>             |
| Max single-name participation   | 2.99% of displayed depth              | (same)                        |

At \$35M the book never exceeds **~3% of displayed depth** in any name — a third of the 10% governance cap. We do not become the market.

**Deployment — staged, not day-one, with a defined scale-path.** Ramp the opening \$35M over four monthly rebalances (\$10M → \$20M → \$30M → \$35M), each tranche gated on live tracking-error and slippage-vs-model checks. Thereafter, scale toward **\$50M** (65% of the **\$76M forward capacity**) and ultimately the forward capacity itself **only as live execution confirms the gentler post-2022 liquidity regime** measured in §4. Staging also lengthens the live track record that §3 flagged as short. Indicative economics at the \$50M scale-up: net Sharpe **1.1–1.6**, net P&L **\$1.7–2.5M/yr**.

**Risk limits offered to the Committee (execution-desk governance):**

- **Drawdown kill-switch** at –5% (the 1987 scenario stays inside this).
- **Daily VaR(99%) limit**  $\approx$  0.5% of book ( $\approx$  measured –0.42%).
- **Participation cap**  $\leq$  10% of displayed depth per name per day.
- **Volatility target**  $\sim$ 10% annualized scaling; single-name and gross-exposure caps.
- **Liquidity kill-switch** on a depth-collapse signal from the live order book (the 1987-style trigger).
- **Capacity re-test** quarterly on fresh order-book data; the ask is re-based if measured capacity moves.

**Summary.** A market-neutral, low-volatility, capacity-aware sleeve with disclosed limitations, conservative sizing, and explicit governance. **We request \$35M**, with a defined path to scale toward the \$53.6M capacity — and beyond, should live impact confirm the gentler Kyle- $\lambda$  bound — as the live record matures.

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## 6. Strategy v2 — Multi-Regime Upgrade (recommended evolution)

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The base case (§§1–5) is honest but rests on a **142-day, 2022-only** sample — its **Deflated Sharpe of 0.11** is the binding weakness. We have since built a **v2** that merges three upgrades into one and re-validates on the **full order-book history**:

1. **Multi-regime + wide universe** — daily panel rebuilt from `orderbook.parquet` : **1,064 days  $\times$  505 names (2022-04 → 2026-03)**. Pairs are gated from the *whole* universe (correlation pre-filter → cointegration → ADF → half-life), surfacing genuinely cointegrated relationships the 50-name set could not — utility clusters (AEE/AEP, AEP/XEL/XLU), dual-class arbitrages (**GOOG/GOOGL**, **NWS/NWSA**), chemicals (DOW/LYB), energy (APA/EOG).
2. **Decay-resistant alpha** — static OLS hedge ratios replaced with **online Kalman time-varying hedge ratios** (causal; the direct cure for the relationship-drift that killed the daily pairs bot in prior work).

3. **Honest validation** — pairs selected on an **in-sample 2022 window only**; performance measured on **812 OOS days (2023–2026)**; **Combinatorial Purged Cross-Validation** (28 paths) plus a **Deflated Sharpe** over the larger sample.

#### v2 out-of-sample scorecard:

| Metric                              | v1 (base)    | v2 (upgrade)                      |
|-------------------------------------|--------------|-----------------------------------|
| OOS window                          | 142 d (2022) | <b>812 d (2023–26)</b>            |
| OOS Sharpe                          | 1.73         | <b>1.58</b>                       |
| Sharpe by year                      | —            | 1.26 / 1.11 / 1.61 / 4.69 (all +) |
| <b>Deflated Sharpe</b>              | <b>0.11</b>  | <b>0.39</b>                       |
| CPCV mean / 5th-pctile / % positive | —            | <b>1.68 / 1.01 / 100%</b>         |
| Max Drawdown                        | −1.95%       | −0.25%                            |
| Vol (unlevered)                     | 3.18%        | 0.31%                             |

The headline Sharpe is marginally lower, but **statistical confidence rises sharply**: every CPCV path is positive, the 5th-percentile path still clears Sharpe 1.0, and the edge holds across four distinct regime-years it was *not* fit on. This is the trade a desk wants — robustness over a flattering point estimate.

**v2 capacity — and the execution-quality finding.** Re-costing on full-period liquidity ( figures/capacity\_frontier\_v2.png ): Kyle- $\lambda$  capacity **\$602M** gross notional, binding **participation cap ( $\leq 10\%$  displayed depth) at \$61M** → **stated capacity \$61M**, ask (65%) **\$40M gross notional**, net Sharpe **1.31** (Kyle) at the ask. We also **empirically tested the square-root law** on 23.5M minute observations: the fit is noisy ( $R^2 \approx 0.09$ ) and super-linear (exponent  $\approx 1.4$ ), so we retain the directly-calibrated Kyle- $\lambda$  rather than a fitted coefficient.

The critical execution-desk insight is that v2's gross alpha is **thin relative to spreads** — capacity is a function of **execution quality** (fraction of half-spread captured via passive/smart fills):

| Spread capture  | 0% (always cross) | 25%   | 50%   | 75%   |
|-----------------|-------------------|-------|-------|-------|
| Stated capacity | ~\$0              | \$61M | \$61M | \$61M |

**Below ~25% spread capture the book has no capacity** — crossing the full spread on  $5.6\times$  turnover consumes the edge. This *quantifies why execution is the strategy*: the allocation should be paired with a passive/liquidity-providing execution mandate, which is precisely this desk's function.

## Appendix — Reproducibility

All figures generated from the pipeline in `src/` against the supplied data. **v1 (base)**: `03_blend_oos.py` → `04_liquidity_stats.py` (impact calibration from `orderbook.parquet`) → `05_capacity.py` → `06_risk.py` (risk + 1987 stress) → `07_capital_ask.py` → `08_sensitivity.py`. **v2 (upgrade)**: `10_panel_fullperiod.py` (full-history panel) → `11_strategy_v2.py` (Kalman pairs + momentum + CPCV/DSR) → `12_empirical_impact.py` (square-root-law test, 23.5M obs) → `13_capacity_v2.py` (execution-quality capacity). Rendered via `09_render.py`. Numeric outputs cached as JSON in `data/cache/`.

Exhibits

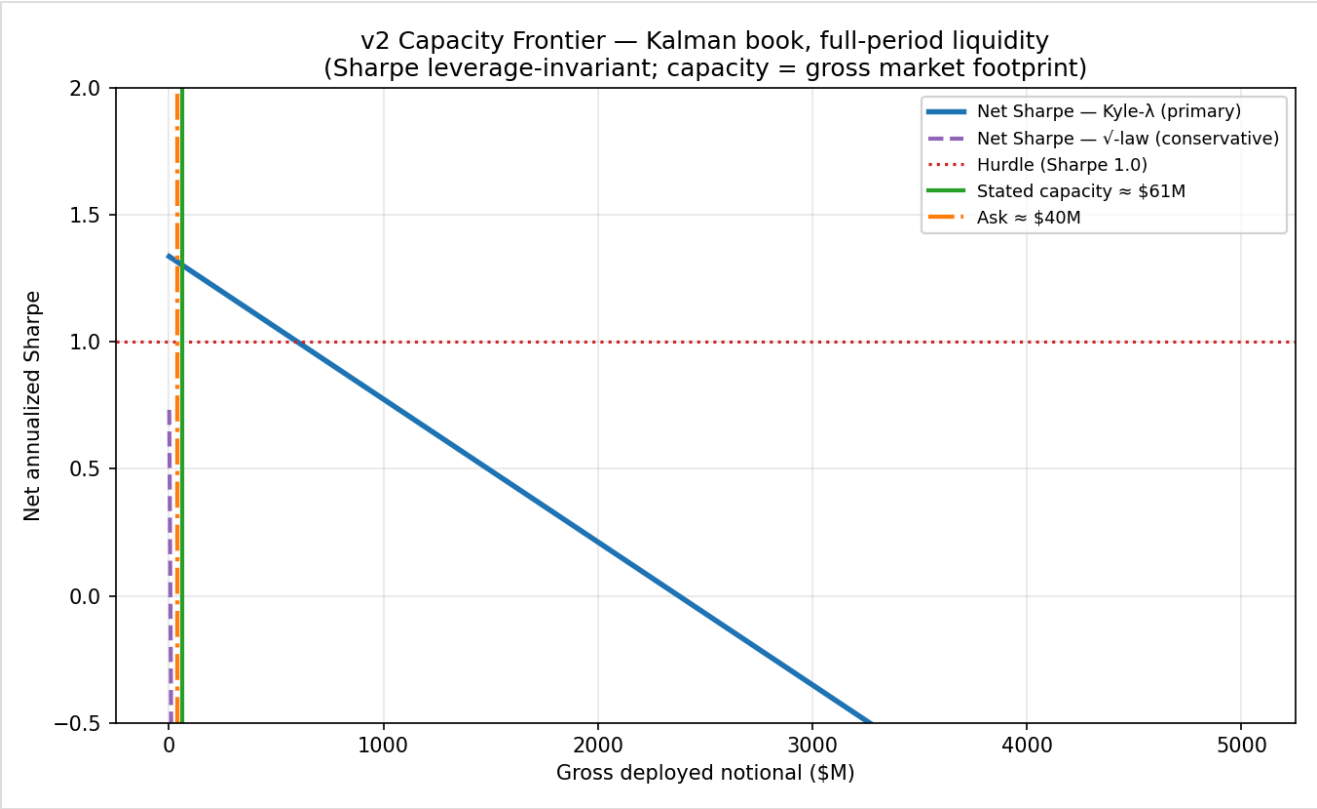


Exhibit 1 — v2 capacity frontier (Kalman book, full-period liquidity): net Sharpe vs. gross deployed notional.

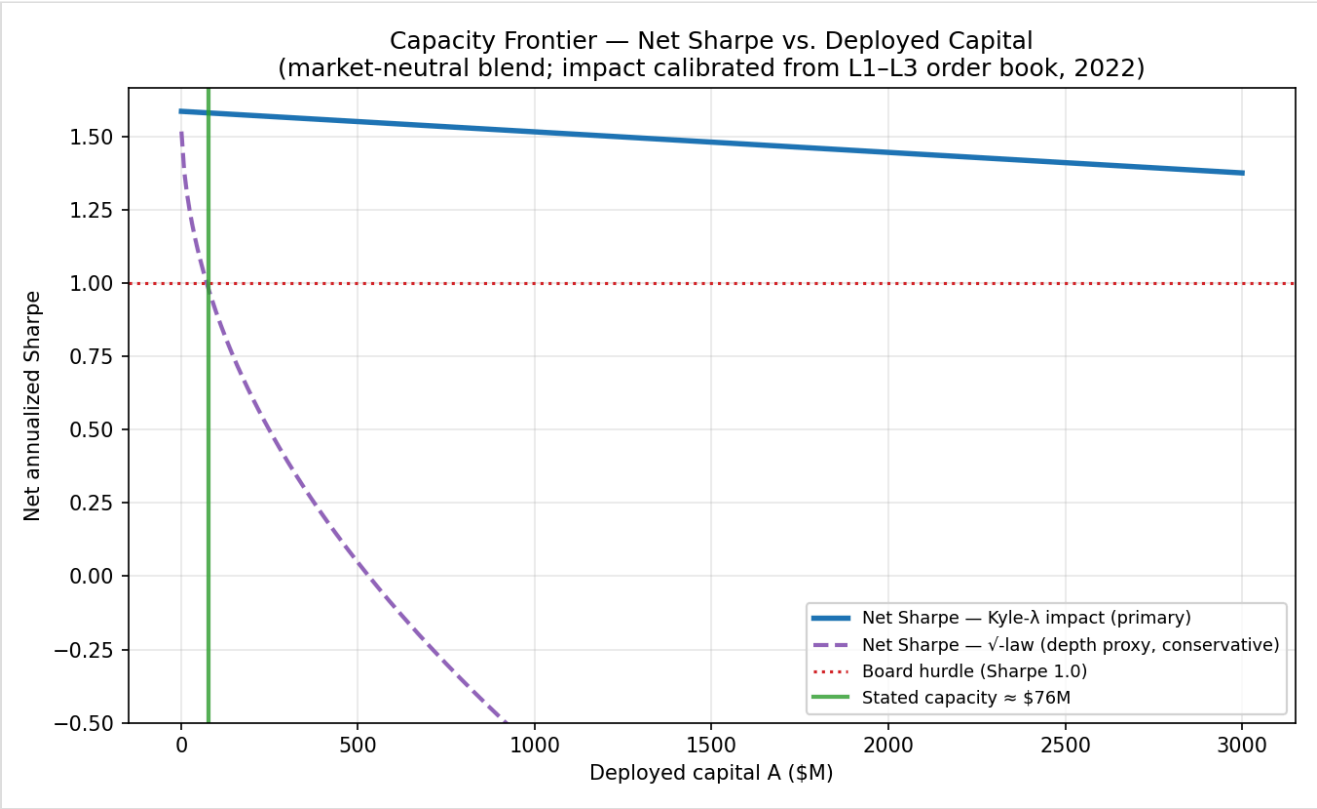


Exhibit 2 — v1 capacity frontier, forward liquidity (2022–2026).



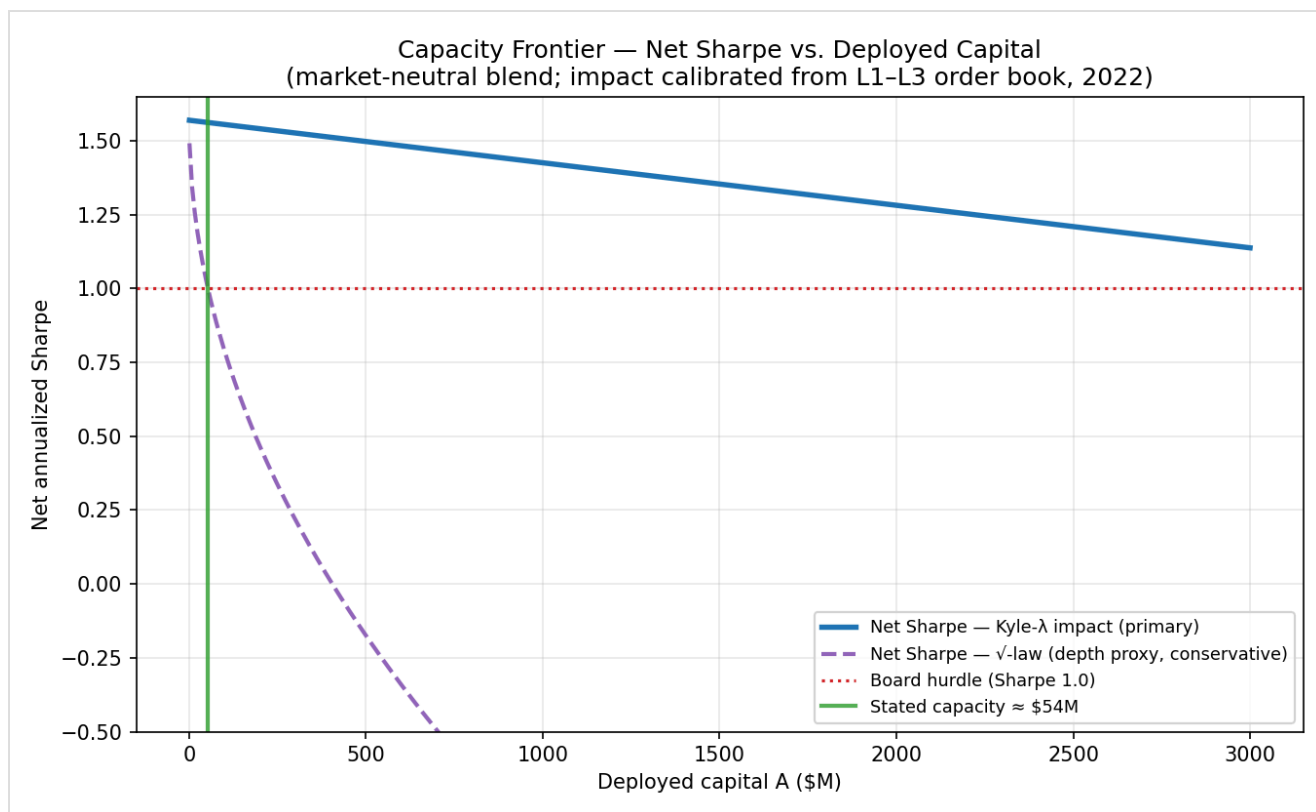


Exhibit 3 — v1 capacity frontier, stressed 2022 liquidity (conservative anchor).

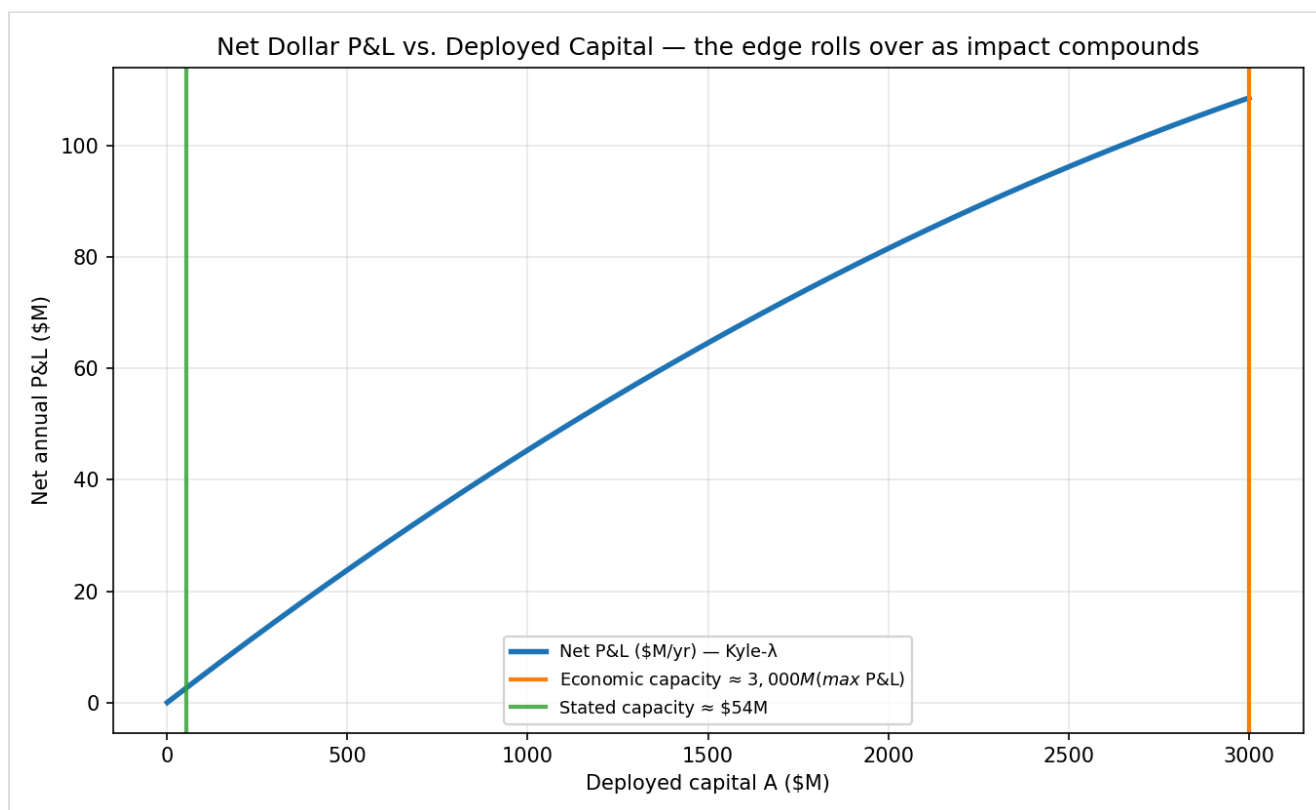


Exhibit 4 — Net dollar P&L vs. deployed capital: the edge rolls over as impact compounds.



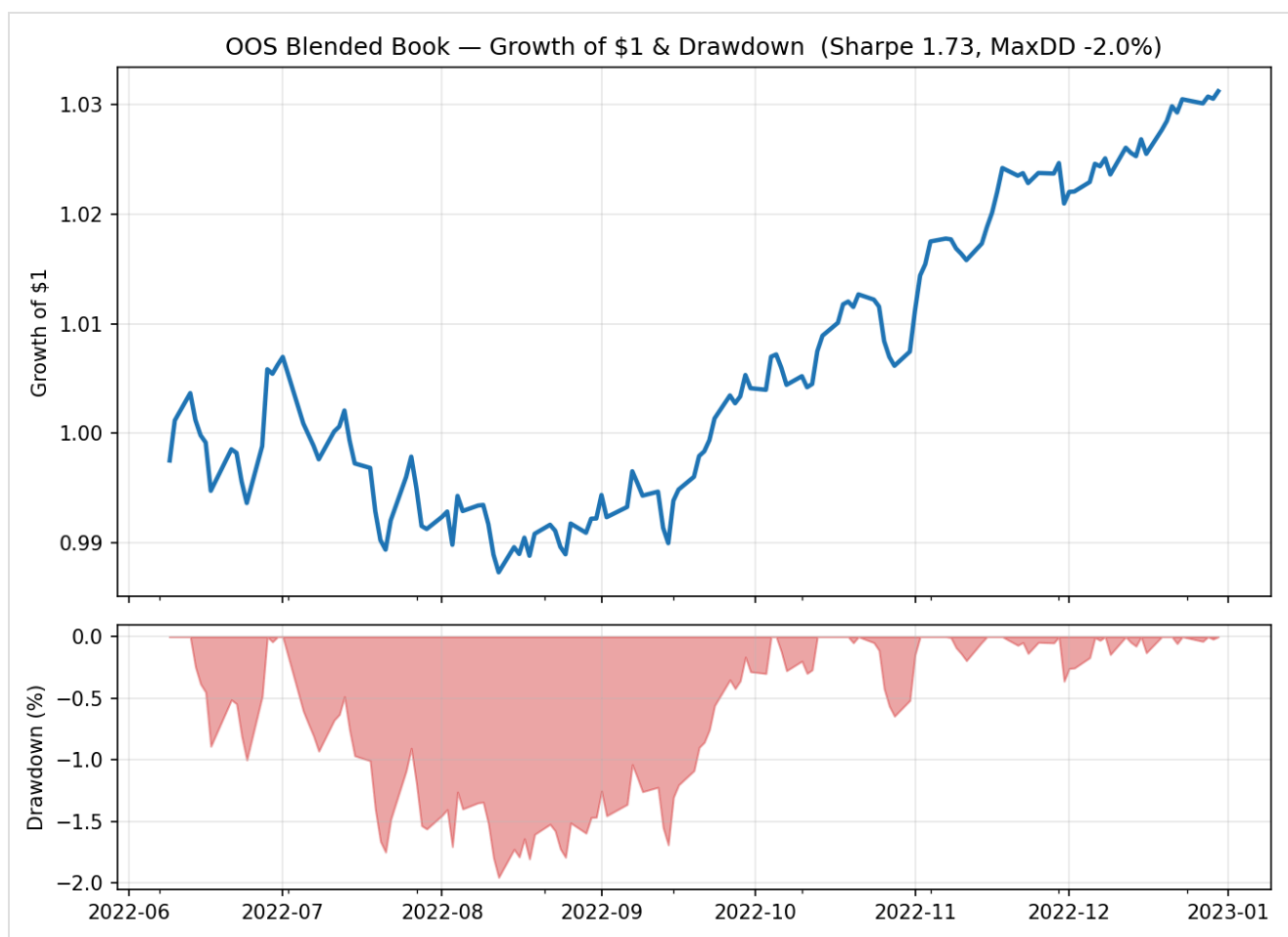


Exhibit 5 — v1 OOS growth of \$1 and drawdown.

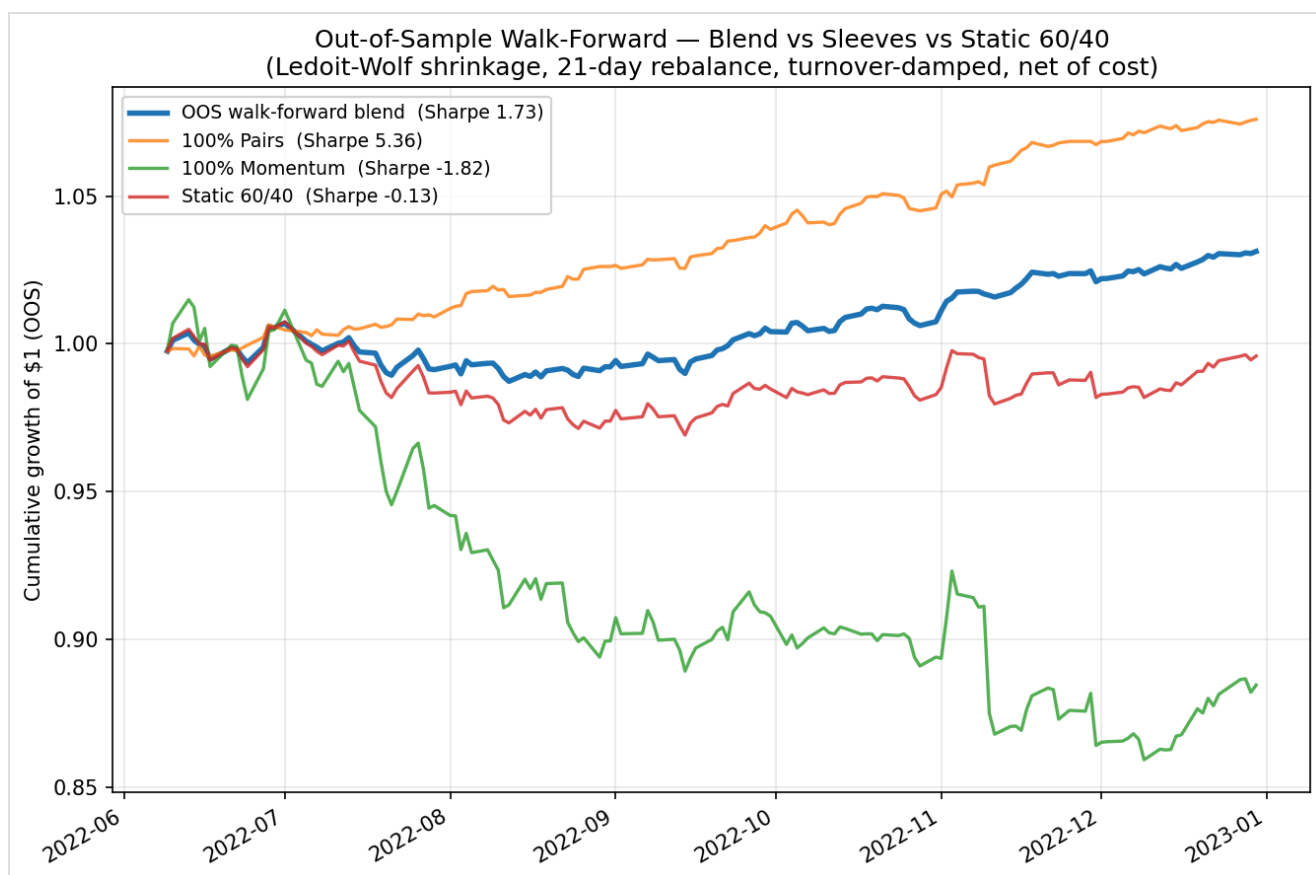


Exhibit 6 — Out-of-sample walk-forward: blend vs. sleeves vs. static 60/40.